

PHASE 3 – PROJECT DESIGN

System Architecture

- Frontend: Streamlit – collects input, shows preview, supports editing and downloads.
- Backend: FastAPI – handles API requests and document-generation orchestration.
- AI Core: Gemini integration – generates structured legal content.
- Formatting Modules: TXT, DOCX and PDF generation.

System Flow

1. User opens the LegalEase Streamlit interface.
2. User enters document type, parties, terms and effective date.
3. Streamlit sends the information to FastAPI.
4. FastAPI validates the request.
5. AI generation component creates the document content.
6. Generated content is returned to the frontend.
7. User previews/edits the content.
8. User downloads TXT, DOCX or PDF.

Main Modules

- Frontend module
- FastAPI/API module
- AI generation module
- Document formatting/export module
- Environment/configuration module

Interface Design

- The Streamlit interface provides fields for document type, parties, terms and dates, together with Generate, Edit and Download functions.
- The generated output is presented as a styled preview.

API Design

- POST /generate is the documented document-generation endpoint.
- The request contains document_type, parties, terms and dates.
- Pydantic is used for request validation.